Conference on Retroviruses and Opportunistic Infections
Denver, Colorado
March 3-6 2024

Back

Doxycycline Post-Exposure Prophylaxis (DoxyPEP): High uptake and Significant Decline in STIs after Clinical Implementation

CROI 2024 March 3-6 Denver

Hyman Scott, MD MPH, Jorge Roman, Matthew Spinelli, Jason Bena, Thiago Torres, and Susan Buchbinder

"The evidence now overwhelmingly supports the use of DoxyPEP for STI prevention and we see benefits of an aggressive rollout to the populations who are most likely to benefit," remarked Hyman Scott, MD, MPH, Medical Director, San Francisco AIDS Foundations. "Despite reservations about widespread adoption, including concerns about antimicrobial resistance, the proactive distribution of DoxyPEP stands poised as a powerful tool to prevent STIs across our communities."

https://www.sfaf.org/collections/breaking-news/san-francisco-aids-foundation-highlights-significant-decline-in-stis-due-to-doxy-pep/

Wide Uptake of DoxyPEP in SF Group, Fast Drop in STI Incidence

CROI 2024 (Conference on Retroviruses and Opportunistic Infections), March 3-6, 2024, Denver

Mark Mascolini

Doxycycline postexposure prophylaxis (DoxyPEP) cut sexually transmitted infection (STI) incidence 60% or more in a 3000-person San Francisco study of men, women, and nonbinary people [1]. Uptake was highest in transgender women, nonbinary people, and cisgender men.

Hyman Scott and colleagues at the San Francisco Department of Public Health and coworkers at other centers noted that rates of STIs (especially syphilis) continue to climb in the United States (especially in men who have sex with men, MSM). Studies of MSM and transgender women show that doxycycline taken after exposure to sexually transmitted chlamydia, syphilis, or gonorrhea cut incidence of these STIs by 50% to 70%. But a single study of DoxyPEP in cisgender women found no benefit with DoxyPEP versus standard care [2]. US national guidelines recommending DoxyPEP were in the works at the time of Scott's presentation at CROI 2024.

The San Francisco Department of Public Health recommended DoxyPEP for at-risk people in October 2022. An STI clinic in San Francisco that sees more than 8000 people for sexual health services began offering DoxyPEP in November 2022. When rolling out DoxyPEP, the clinic planned to evaluate STI incidence trends with this postexposure strategy. The clinic considers recommending DoxyPEP for gay, bisexual, or heterosexual men, as well as for transgender and gender-nonconforming people, while use in cisgender women remains under study.

The clinic explains how DoxyPEP works in plain language: "By taking two pills (100 mg each) of the antibiotic doxycycline ("Doxy") within 24 to 72 hours of fucking or sucking as the top, bottom, or both, you can stop the spread of STIs. One dose of Doxy PEP will cover all the sex you'll have in a day . . . Don't take more than two pills (one dose) in a 24-hour period."

This analysis included all active San Francisco AIDS Foundation HIV preexposure prophylaxis (PrEP) clients from June 2022 through September 2023, all of whom were offered DoxyPEP at routine visits starting on November 30, 2022. Researchers assessed two cohortsâ€" the DoxyPEP cohort, who started PEP any time after November 30, 2022, and the non-DoxyPEP cohort, who never started the preventive strategy during the study period. Analysis yielded (1) incidence rate ratios (IRR) comparing the pre-DoxyPEP period (June 2022 through November 2023) and the post-DoxyPEP period (30 days after starting DoxyPEP), and (2) a controlled interrupted time series [3] using mixed effects Poisson regression.

Among 3081 total PrEP clients, 1209 (39%) started DoxyPEP, including 37% of 509 Asians, 37% of 126 blacks, 43% of 723 Hispanic people, 41% of 408 multiracial people, and 37% of 1095 whites. Overall, 45% declined DoxyPEP or had an unknown DoxyPEP uptake status. Uptake rates did not vary much by age, ranging from 34% to 43% in 5 age groups under 60 and falling slightly to 27% in people 60 or older. All 14 cisgender women offered DoxyPEP declined, while 25% of 32 transgender men, 39% of 2763 cisgender men, 42% of 77 transgender women, and 44% of 188 nonbinary people opted to try DoxyPEP.

Incidence of any STI (chlamydia, syphilis, or gonorrhea at any site) fell about 60% in the pre-post analysis (IRR 0.42, 95% confidence

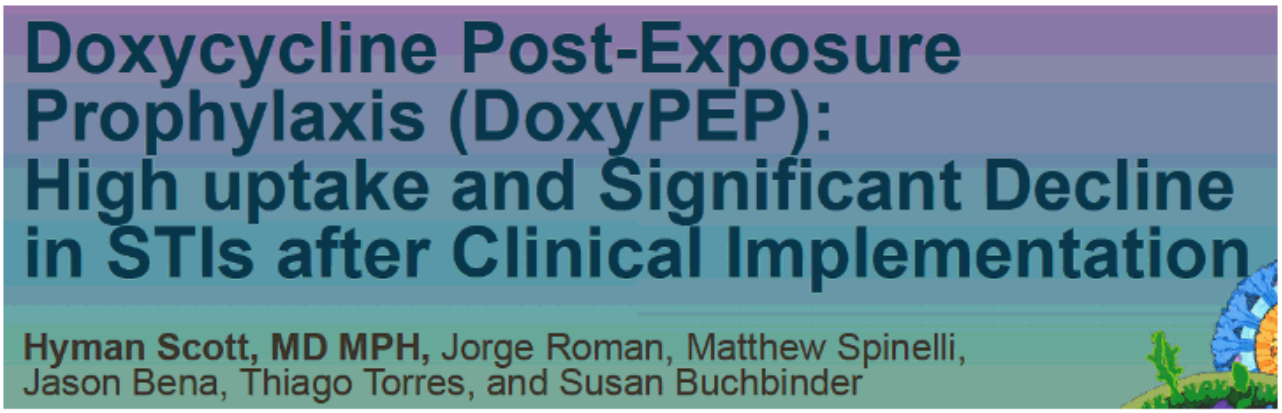
interval [CI] 0.24 to 0.74, P = 0.003), while incidence of syphilis plunged almost 80% (IRR 0.22, 95% CI 0.09 to 0.54, P = 0.001) and incidence of chlamydia waned almost 70% (IRR 0.33, 95% CI 0.23 to 0.46, P < 0.001). Gonorrhea incidence slipped about 10%, a nonsignificant change (IRR 0.89, 95% CI 0.69 to 1.15, P = 0.383). But gonorrhea incidence did appear to be falling over time with DoxyPEP in the interrupted time series analysis.

For any STI, interrupted time series analysis showed a one-third decline for post- versus pre-DoxyPEP (0.67, 95% CI 0.46 to 0.96, P = 0.03). Interrupted time series also confirmed significant ebbs in incidence of chlamydia (P = 0.021) and gonorrhea (P = 0.003).

Scott and coworkers believe their findings reflect "strong demand [for DoxyPEP] among potential users." Uptake proved highest among people with STIs, cisgender men, and transgender women. STI incidence fell swiftly after DoxyPEP began, a finding "demonstrating high impact of this intervention in a real-world setting."

References

1. Scott H, Roman J, Spinelli MA, et al. Doxycycline PEP: High uptake and significant decline in STIs after clinical implementation. CROI 2024 (Conference on Retroviruses and Opportunistic Infections), March 3-6, 2024, Denver. Abstract 126.
2. Stewart J, Oware K, Donnell D, et al. Doxycycline prophylaxis to prevent sexually transmitted infections in women. N Engl J Med. 2023;389:2331-2340. DOI: 10.1056/NEJMoa2304007. https://www.nejm.org/doi/10.1056/NEJMoa2304007 3. English P. Package 'its.analysis.' cran.r-project.org. October 13, 2022. https://cran.r-project.org/web/packages/its.analysis/its.analysis.pdf



Conclusions

- DoxyPEP uptake was high reflecting strong demand among potential users.
  - Uptake was highest among those with STIs; cisgender men and transgender women; and appears to be equitable across race/ethnicity
- Overall STI incidence declined rapidly after implementation demonstrating high impact of this intervention in a real-world setting.
- No impact of DoxyPEP on gonorrhea incidence in the pre- post-DoxyPEP analysis
  - Incidence does appear to be declining over time among DoxyPEP users based on ITS analysis
- Continued evaluation of uptake, adherence, and impact on bacterial STIs will be essential with expanded DoxyPEP implementation.

Background

- Bacterial STI cases have continued to increase in the US, especially among cisgender men who have sex with men (MSM)
  - Syphilis cases continue to increase faster than other bacterial STIs
- DoxyPEP has been shown to reduce chlamydia and syphilis by ~70%, and gonorrhea by ~ 50% in several studies among MSM and transgender woman (TGW).
- A single study of DoxyPEP among cisgender woman failed to show efficacy.
  - Doxycycline levels were only detected in 29% of a random subset of participants
- Local US guidelines have recommended DoxyPEP use for certain populations, and national guidelines are pending (as of 2/2024).

Molina JM et al, Lancet Infect Dis 2018; Luetkemeyer A et al, NEJM 2023; Molina JM CROI 2023, Stewart J NEJM 2023, https://www.cdc.gov/std/statistics/infographic.htm

SFDPH Recommended DoxyPEP in 10/2022

Health Update

Doxycycline Post-Exposure Prophylaxis Reduces Incidence of Sexually Transmitted Infections

October 20, 2022

Magnet Clinic @ Strut - San Francisco AIDS Foundation

- Large sexual health clinic located in the Castro Neighborhood
  - >8,000 clients seen annually for sexual health services
  - ~3,000 active PrEP users
- DoxyPEP rollout started in November 2022
- Client-centered advertising in clinic; social media posts; and discussion at all routine PrEP visits.
- **Planned evaluation of STI incidence among PrEP users following DoxyPEP implementation.**

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Doxy-PEP Client & Community Education

SAN FRANCISCO AIDS FOUNDATION

WHO IS IT FOR?

Doxy PEP is for gay, bi, or hetero cis guys, plus trans, non-binary, and TGNCI+ folx* who want to protect themselves from gonorrhea, chlamydia, and syphilis.

At this time, it's under study with cis women. We will let you know when we know more about how well this works for cis women.

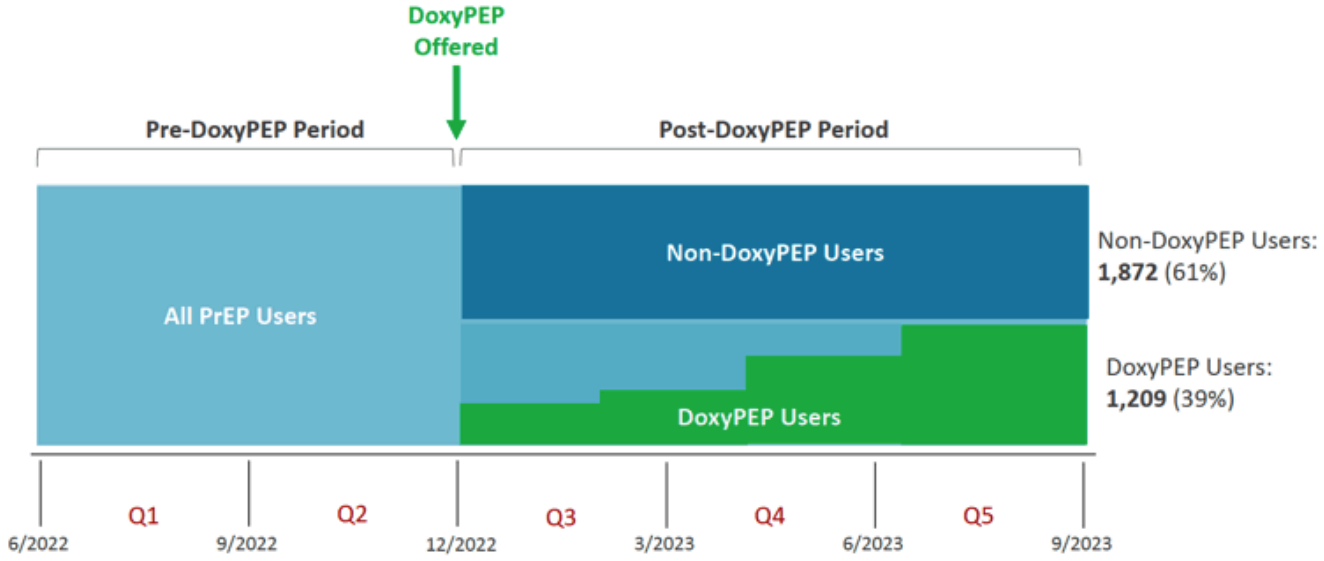
HOW DOES IT WORK?

By taking two pills (100 mg each) of the antibiotic doxycycline ("Doxy") within 24 to 72 hours of fucking or sucking as the top, bottom, or both, you can stop the spread of STIs. One dose of Doxy PEP will cover all the sex you'll have in a day — so make Doxy PEP your last load of the night. Don't take more than two pills (one dose) in a 24 hour period.

Population and Methods

- All active SFAF PrEP clients from 6/2022 through 9/2023 were included in this analysis
  - DoxyPEP was offered to all PrEP clients at routine visit starting on 11/30/2022
- Two exclusive cohorts:
  - DoxyPEP: Clients who started DoxyPEP at any time after 11/30/2022
  - Non-DoxyPEP: Clients who never started DoxyPEP during the study period
- STI Incidence was evaluated per quarter
  - Pre-DoxyPEP Period: 6/2022 through 11/2023
  - Post-DoxyPEP Period: 30 days after DoxyPEP initiation (among DoxyPEP users).
- Analysis:
  - Incidence Rate Ratios (IRR) comparing pre- post-DoxyPEP among DoxyPEP users, and
  - Controlled Interrupted Time Series (ITS) using mixed-effects Poisson regression

DoxyPEP Timeline



Race/Ethnicity

Race /Ethnicity	Total PrEP Clients N=3,081 n	DoxyPEP Uptake n=1,209 %
American Indian or Alaska Native	9	56%
Asian	509	37%
Black or African American	126	37%
Hispanic or Latinx	723	43%
Multi-Racial	408	41%
Native Hawaiian or Pacific Islander	16	25%
White	1,095	37%
Declined/Other/Unknown	195	45%

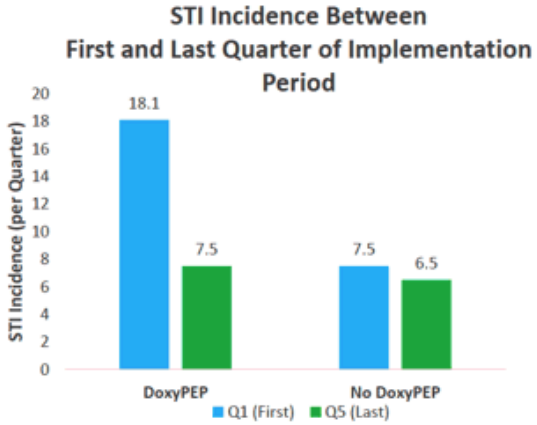
Age

Age (years)	Total PrEP Clients N=3,081 n	DoxyPEP Uptake n=1,209 %
18-24	223	38%
25-29	636	40%
30-39	1,299	43%
40-49	535	36%
50-59	256	34%
60+	132	27%

Gender Identity

Gender Identity	Total PrEP Clients N=3,081 n	DoxyPEP Uptake n=1,209 %
Cisgender Man	2,763	39%
Cisgender Woman	14	0%
Transgender woman	77	42%
Transgender man	32	25%
Non-binary	188	44%
Declined/Other/Unknown	7	57%

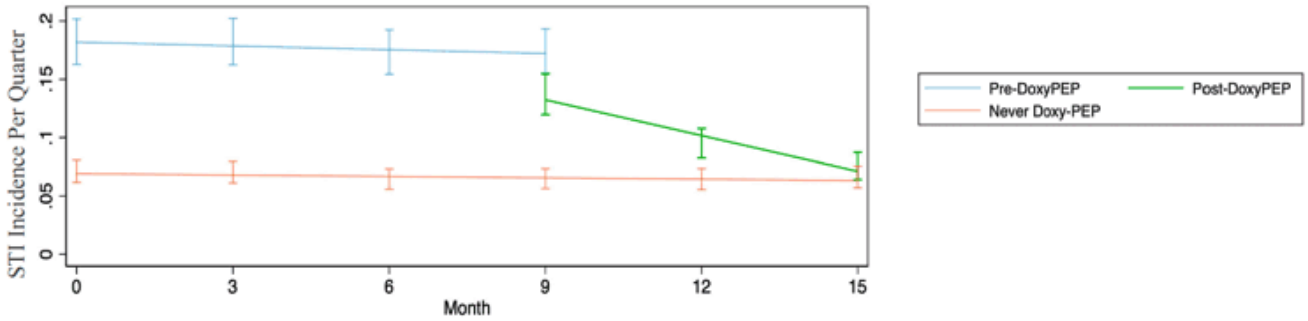
STI Incidence among DoxyPEP Users (Pre-Post Analysis)



	IRR	95% CI	p-value
Any STI	**0.42**	**0.24 - 0.74**	**0.003**
Chlamydia	**0.33**	**0.23 - 0.46**	**<0.001**
Syphilis	**0.22**	**0.09 - 0.54**	**0.001**
Gonorrhea	0.89	0.69 - 1.15	0.383

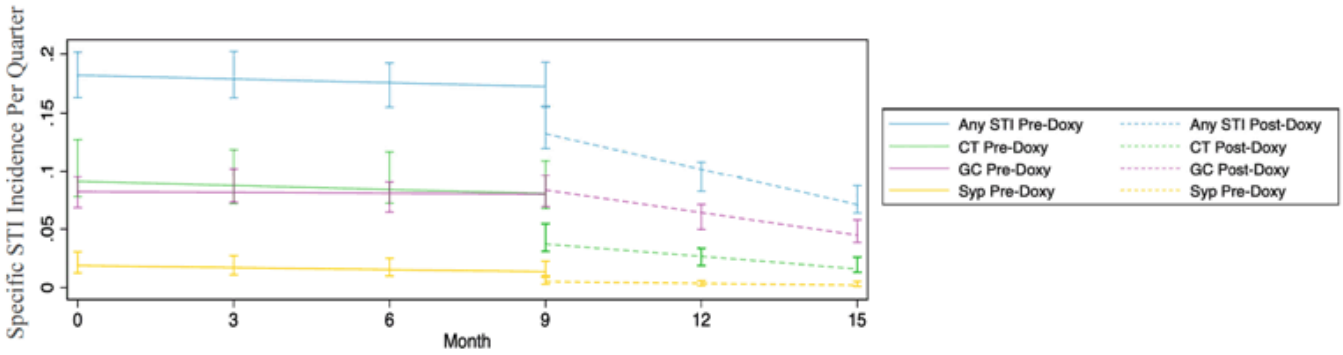
Any STI: Chlamydia, Syphilis, or Gonorrhea at any site.

Interrupted Time Series: Any STI Pre- Post-DoxyPEP



ITS: Any STI among DoxyPEP Users: 0.67, 95% CI: 0.46-0.96; p<0.03

Interrupted Time Series: STI Incidence Pre- Post-DoxyPEP



ITS: Any STI: p=0.32; Chlamydia: p=0.021; Gonorrhea: p=0.003; Syphilis: p=0.360

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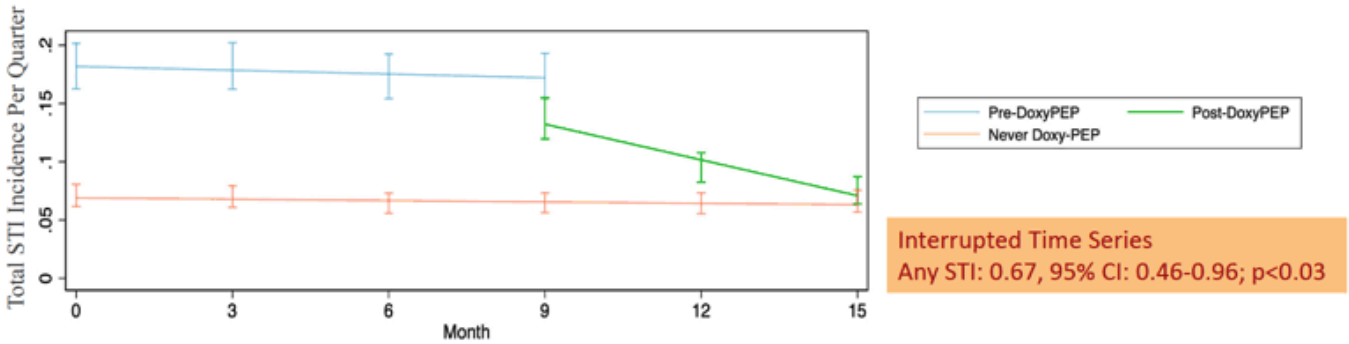
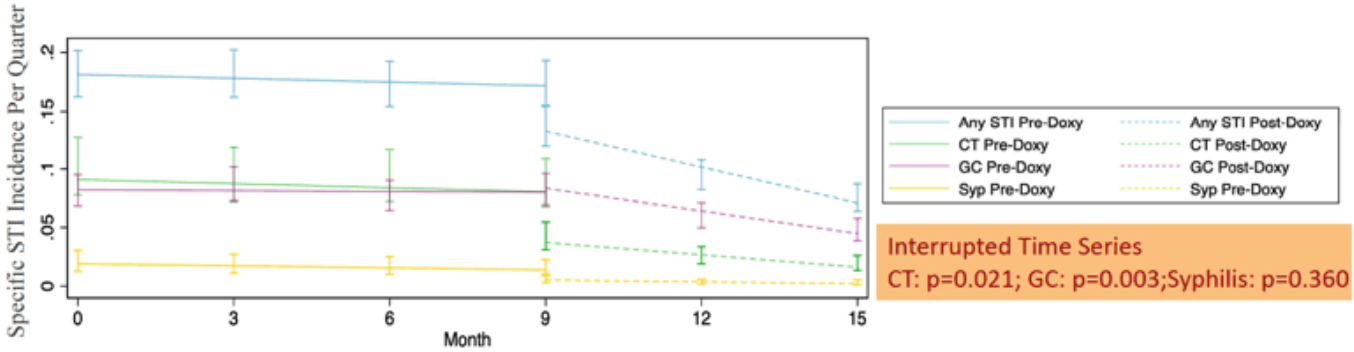


Interrupted Time Series
Any STI: 0.67, 95% CI: 0.46-0.96; p<0.03



Interrupted Time Series
CT: p=0.021; GC: p=0.003;Syphilis: p=0.360

Race/Ethnicity

Race /Ethnicity	Non-DoxyPEP n=1,872 %	DoxyPEP n=1,209 %	Total PrEP Clients N=3,081 %
American Indian or Alaska Native	0.2	0.4	0.3
Asian	17.2	15.5	16.5
Black or African American	4.3	3.8	4.1
Hispanic or Latinx	22.2	25.5	23.5
Multi-Racial	12.8	14.0	13.2
Native Hawaiian or Pacific Islander	0.6	0.3	0.5
White	37.0	33.3	35.5
Declined/Other/Unknown	5.7	7.3	6.3

Gender Identity

Gender Identity	Non-DoxyPEP n=1,872 %	DoxyPEP n=1,209 %	Total PrEP Clients N=3,081 %
Cisgender Man	89.8	89.5	89.7
Cisgender Woman	0.7	0	0.5
Transgender woman	2.4	2.6	2.5
Transgender man	1.3	0.7	1.0
Non-binary	5.6	6.9	6.1
Declined/Other/Unknown	0.2	0.3	0.2

Age

Age Category (years)	Non-DoxyPEP n=1,872 %	DoxyPEP n=1,209 %	Total PrEP Clients N=3,081 %
18-24	7.4	7.0	7.2
25-29	20.6	20.8	20.6
30-39	39.6	46.2	42.2
40-49	18.3	15.9	17.4
50-59	9.0	7.2	8.3
60+	5.1	3.0	4.3

Acknowledgments

- Jorge Roman (SFAF)
- Jason Bena (SFAF)
- Thiago Torres (FioCruz)
- Matthew Spinelli (UCSF)
- Susan Buchbinder (SFDPH)